While VQE is meant for calculating the ground state energy of a Hamiltonian, it is also of interest to higher energy states, also known as "excited states" [24–26]. Computing excited states provides insight into the properties of quantum systems. In quantum chemistry, determining excited states helps to understand chemical reaction mechanisms, including the absorption spectra and photochemistry of molecules [27]. In materials science, excited states can reveal information about electronic properties such as conductivity and magnetism, which are vital to designing new materials with targeted functionalities [28]. Calculating these excited states, as opposed to only finding the ground state, provides broader information on molecules of interest, and poses a greater technical challenge due to the requirement of maintaining orthogonality among calculated states.

Motivation and our contributions. This work builds upon a classical collaborative computation model, the EigenGame, presented in [29], to compute excited states in a noisy quantum environment. Unlike existing approaches that rely on deflation to calculate successive states [30–35], our method allows such a computation via a regularized objective inspired by game-theoretic ideas. Our primary contributions are:

- *Algorithmic Framework*: We propose a VQE meta-algorithm that changes the computational dynamics to find excited states. Using a modified quantum objective, our approach avoids deflation steps, diverging from traditional deflation-based computation models.
- *Theoretical Component*: We provide a theory that validates the convergence properties of our algorithm. By formalizing the interaction between error calculations –due to noisy gradients and DFO routines– and convergence rates, we establish a theory for VQE targeting excited states.
- *Empirical Validation*: The current implementation shows that our approach leads to favorable performance compared to baseline algorithms in terms of either convergence rate or accuracy.

2. Background

Notation. Matrices are denoted by uppercase letters, such as $A \in \mathbb{C}^{p\times p}$, while lowercase letters, like $b \in \mathbb{C}^p$, represent vectors. Scalars are distinguished based on the context. The Euclidean ℓ_2-norm is symbolized by $\|\cdot\|_2$. The *qubit* is the basic unit, analogous to the bit in classical computing. A qubit's state is expressed using the Dirac (bra-ket) notation; a single qubit state $|\psi\rangle \in \mathbb{C}^2$ is a linear combination of the basis states $|0\rangle = [1\ 0]^\top$ and $|1\rangle = [0\ 1]^\top$ in $\mathbb{C}^2$. For example, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $\alpha, \beta \in \mathbb{C}$ are complex amplitudes. These amplitudes encode the probabilities that the qubit's state collapses to $|0\rangle$ or $|1\rangle$, satisfying the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

The notation $|\cdot\rangle$ represents a column vector (referred to as a *ket*), and $\langle\cdot|$ denotes its conjugate transpose (*bra*). This is a convenient notation that allows an inner product to be expressed by $\langle x|y\rangle$ and an outer product to be represented by $|x\rangle\langle y|$, with $\langle x|x\rangle = 1$ and $\langle x|y\rangle = 0$, for $x \perp y$. A separable q-qubit state, residing in a q-qubit Hilbert space, is the Kronecker product of q individual qubit states, represented as $\mathcal{H} = \otimes_{i=1}^{q}\mathbb{C}^2 \cong \mathbb{C}^{2^q}$. It is expected to equate $2^q = p$. Quantum states are manipulated by quantum gates, which are unitary matrices acting on state vectors. A single-qubit gate $U \in \mathbb{C}^{2\times 2}$, for example, transforms a state $|\psi\rangle$ into $U|\psi\rangle$, adjusting the state's probability amplitudes according to the operation defined by U. An ideal quantum gate U is defined as a unitary matrix, where $U^\dagger U = UU^\dagger = I$ allows rotation effects on $|\psi\rangle$ to preserve the normalization condition of the amplitude.

Principal component analysis and VQE. PCA has been studied as a way of finding the representative components from matrix data [36, 37]. This unsupervised learning problem effectively looks for the top (or bottom) principal component of a data matrix calculated via:

$$\max_{v\in\mathbb{R}^p:\|v\|_2=1} / \min\ v^\top M v, \tag{1}$$

where $M = \frac{1}{n}X^\top X \in \mathbb{R}^{p\times p}$ is often the covariance matrix of a dataset $X \in \mathbb{R}^{n\times p}$ with usually centered, normalized rows. For our discussion, assume that M is a real symmetric matrix with eigenvalues and eigenvectors $\{\lambda_i^\star, v_i^\star\}_{i=1}^p$, satisfying $\lambda_1^\star \geq \lambda_2^\star \geq \cdots \geq \lambda_p^\star \geq 0$. Then, $v_1^\star$ corresponds to the vector that maximizes (1) with objective value $v_1^{\star\top} M v_1^\star = \lambda_1^\star$, and $v_p^\star$ is the vector that minimizes (1) with objective value $v_p^{\star\top} M v_p^\star = \lambda_p^\star$.